

Integrated Radar Sensing and Communications: convergence, co-design and learning

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Abstract

Integrated sensing and communications (ISAC) systems operate within shared, congested, or even contested spectrum, aiming to deliver high performance in both wireless communications and radio frequency (RF) sensing. Communications and sensing functionalities are co-designed for mutual benefit. ISAC systems share hardware and antenna resources, use joint waveforms, and function within shared spectra and exchange awareness about their radio environments to jointly optimize performance. ISAC technology has opened genuinely new lines of research and development in emerging 6G systems rather than merely being an evolution of 5G. ISAC takes advantage of the ongoing parallel convergence of modern radar and multi-antenna communications technologies. Sensing may be a service, can aid communications, and conversely, communications may aid sensing. ISAC is expected to provide unprecedented quality of experience towards reliable, resilient, and ubiquitous connectivity, beyond the high data-rate transmission of bits. This talk focuses on various waveform design, signal processing, optimization, adaptation and reinforcement learning techniques employed to achieve desired sensing and communications performances in multicarrier (MC) and multiantenna ISAC systems. Examples on waveform design, optimization, resource allocation in multi-user and multi target scenarios are provided.

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1 Bio

Visa Koivunen (IEEE Fellow, EURASIP Fellow, AAIA Fellow) received the D.Sc. degree (Hons.) in electrical engineering from the Department of Electrical Engineering, University of Oulu. He was a Visiting Researcher with the University of Pennsylvania, Philadelphia, USA, from 1991 to 1995. Since 1999 he has been a Full Professor of signal processing with Aalto University (formerly Helsinki UT), Finland and Aalto Distinguished Professor since 2020. He was an Academy Professor in 2010–14. From 2003 to 2006, he was an Adjunct Full Professor with the University of Pennsylvania, USA. He

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Figure 1: Professor Visa Koivunen

has spent two full sabbaticals and mini-sabbaticals each year with Princeton University. On his sabbatical year from 2022 to 2023, he was a Visiting Professor with EPFL, Lausanne, Switzerland.

His research interests include statistical signal processing, wireless communications, radar, multisensor systems, data science, and machine learning. He was awarded the 2015 European Association for Signal Processing (EURASIP) Technical Achievement Award for fundamental contributions to statistical signal processing and its applications in wireless communications, radar, and related fields. He has co-authored multiple papers receiving the best paper award including the IEEE SP Society Best Paper Award for the year 2007 (with J. Eriksson) and 2017 (with Zoubir, Muma, and Chakhchouk). He has served on the editorial board for The Proceedings of the IEEE, and IEEE Signal Processing Magazine among other journals.